



Augmented Reality Workshop

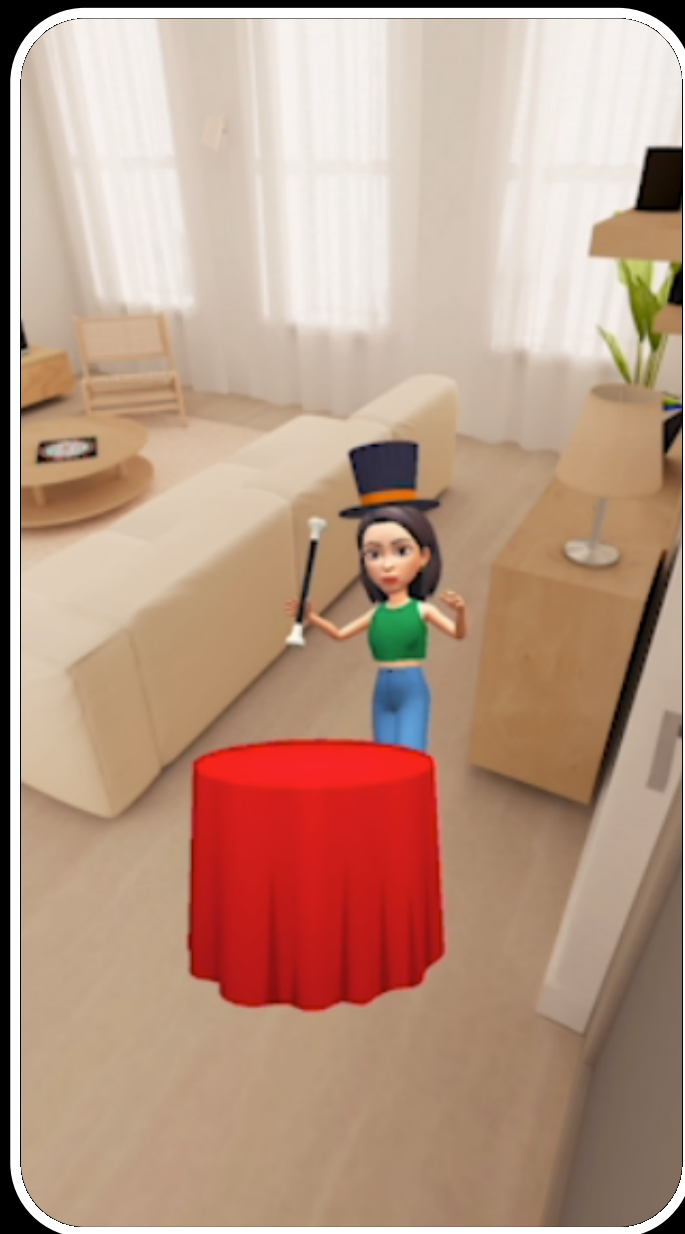
Bitmoji Magician

Helper Document / Snap Chat – Lens Studio Project

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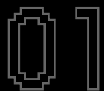
Last Updated: January 2025

This lens studio project aims to create an augmented reality experience for snapchat with an animated bitmoji. When the effect is loaded, and while looking through the rear camera, props are attached, or digital assets, to a Bitmoji avatar.



This project walks through these steps in the context of creating a magician Bitmoji Lens, however, the same process will apply to any theme.

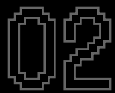
We will be setting up the account, gather all the downloadable resources, prepare some assets. Install and setup lens studio to start a new project.



Download the project files

[↪ Link](#)

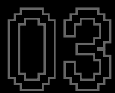
Download the project files for animated dancing bitmoji from github repository. Click on the green "Code" button to download.



Snapchat account

[↪ Link](#)

Go to the official Snapchat website and create a Snapchat account, then login.



Download lens studio

[↪ Link](#)

Downloading the latest lens studio available [5.6 or later]



Open lens studio 5.6+

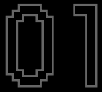
Open the lens studio. Login to your snapchat account.



New Project

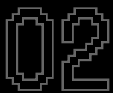
Create a blank project, not templates. Save the project with "ctrl+s" / "cmd+s" to you desired location.

We'll add the necessary assets and begin configuring the project.



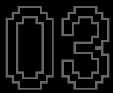
Adding assets from library

- Open the assets library [found above the "scene hierarchy"].
 - Search for "Magician Pack", "Bitmoji Joint Attachment" components and add them.
-



Adding assets to scene hierarchy

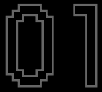
Add the "BitmojiMagician__PLACE_IN_OBJECTS_PANEL" into the scene from asset browser > Magician Pack



Creating prefabs

- Expand the "BitmojiMagician__PLACE_IN_OBJECTS_PANEL" object.
 - Create prefabs for both "baton_binding" and "hat_magic_binding", by dragging the joint from scene and dropping it in the asset browser.
-

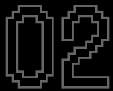
We will generate animations with AI and add them to the project.



GenAI Suite

- Select the "Try Easy Lens" from the top-right menu, which will open up the GenAI suite.
- Select "Bitmoji Animation".
- Import* the animation called, "Excited Reveal", and click on "Import to Project".

*After clicking the import button, the bitmoji will be automatically added to the scene with the selected animation. This may take a while depending on the system specifications. You can choose any animation you like, but for the sake of uniformity, we are going with excited reveal.



Bitmoji animation controller

Select the "BitmojiAniamtionController". This controller will be automatically added with bitmoji animation. On the inspector panel;

- Autoplay: FALSE
 - Playback Mode: Single
-



Bitmoji joint attachment

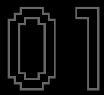
- Select the bitmoji joint attachment object.
 - Drag and drop the "Bitmoji" object to the bitmoji 3d slot.
 - Joint: Right Hand
 - Drag and drop the "baton_binding" prefab from the assets browser to the prefab slot.
-



Bitmoji

- Select the bitmoji object and enable the "is Animated" option.
 - Drag and drop the "BitmojiAnimationaController" to the animation player slot.
-

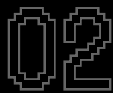
We'll start attaching the baton and hat objects to the bitmoji character.



Camera Object

- Select the "Camera" object.
- Add a "Device Tracking" component*.

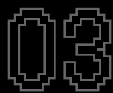
* We are adding the device tracking only to navigate around the scene while we are working. Once we are done, we will remove this device tracking component from the scene.



Simulator

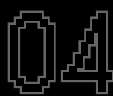
- Change to the rear camera.
- Change to interactive mode.
- Choose the plane* option to view.

* To navigate inside the simulator, hold the right mouse button and w to move forward. RMB +S for backward. RMB + A for left. RMB + D for right. RMB + Q for downward. RMB + E for upward. Hold the RMB and look around for orbiting.



Bitmoji joint attachment {Baton}

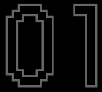
- Rename the object to "Bitmoji Join Attachment – Baton"
 - To offset the prefabs, we can use the position offset, scale and rotation offset.
 - Position offset: -0.02 | 0.11 | 0.10
 - Rotation offset: 25.0 | 25.0 | 0.0
 - Autoplay: TRUE
-



Bitmoji joint attachment {Head}

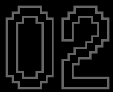
- Duplicate the "Bitmoji Join Attachment – Baton"
- Rename to "Bitmoji Join Attachment – Head"
- Joint: Head
- Drag and drop the "hat_magic_binding" prefab from the assets browser to the prefab slot.
- To offset the prefabs, we can use the position offset, scale and rotation offset.
 - Position offset: 0.15 | 0.11 | -0.01
 - Rotation offset: 180.0 | 0.0 | 0.0
- Autoplay: TRUE

We'll enhance the interactivity and realism.



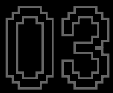
Camera Object

Remove the "Device Tracking" component from the camera object.



Scene hierarchy

- Move the "table_magic_binding" out of its parent object.
 - Delete the parent "BitmojiMagician__PLACE_IN_OBJECTS_PANEL" object.
 - Change the position of "table_magic_binding" to "-1.5 | -12 | 16"
-



Adding assets from library

- Add the asset "Mark the Monster" to the scene.
-



Mark the monster

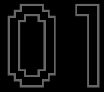
- Drag and drop Mark The Monster onto the table object so that the table is the parent of the MarkTheMonster object.
 - Adjust the position so that he is standing directly on the table.
 - Set the MarkTheMonster SceneObject as inactive in the scene hierarchy.
-



Mark the monster

- Add the Behavior Script to your scene through the Hierarchy panel.
 - Once added, set the trigger to On Start and add a delay time so that this will trigger after 1.5 seconds from when the Lens starts.
 - Set the Response Type to Set Enabled and set the Target to MarkTheMonster.
 - Now Mark The Monster will appear before our eyes after 1.5 seconds.
-

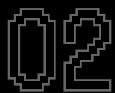
We'll test the effect on our mobile device and upload it to Snapchat for global use.



Preview lens

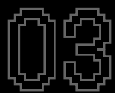
- On the top right of the screen near the publish button, you can choose to pair with your mobile device using your snapchat account.
- Select "Pair new Snapchat Account" and a SnapQR* will be shown.
- Open your snapchat and point the camera to the SnapQR.
- You will receive the prompt to continue the pairing process.
- Once paired the lens will send to your device.

*SnapQR can only be scanned using a snapchat camera. Normal QR code readers won't work with SnapQR.



Automating lens previews

- Click on Lens Studio on top left and select "Preference".
- Send to Device > Send on Project Save: Snapchat Only.



Publish

- On the top-right, select "Publish" to open the project settings window.
- You can add or change the Lens Name, Platforms, Previews and Icons.
- Once customized, you can again select "Publish".
- That will guide you to the snapchat website. Follow the instructions and your effect will be published in a while.



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